Human TFEB in HeLa Cells: Controlled Replication and a Test of C-Terminal Regulation

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Abstract

Transcription factor EB (TFEB) connects nutrient signaling to lysosomal and autophagy-related gene expression. This proposal reproduces nutrient- and Torin1-dependent TFEB redistribution in HeLa cells with a newly designed reporter, then tests how the C-terminal S466/S467/S469 cluster controls target-gene RNA. The WT reporter is 6,134 bp, encodes 722 residues and has a theoretical unmodified mass of 80.42 kDa; it is built by HindIII/BamHI directional cloning into pEGFP-N1 and accepted after bidirectional Sanger sequencing. EGFP-only, S211A, S142A and S142A/S211A controls support matched starvation, Torin1 and refeeding experiments, backed by endogenous TFEB measurements, expression titration, RPTOR/RICTOR RNAi and TFEB loss-and-rescue. The research extension compares WT, S466A/S467A/S469A (3A) and S466T/S467T/S469T (3T) with or without the linker–EGFP fusion under vehicle, Torin1 or the AMPK activator MK-8722; five additional sequence-defined constructs complete the six-construct matrix, and two linker-only constructs attribute any architecture effect to the linker or to EGFP. The primary endpoint is MCOLN1 RNA at 6 h, analyzed as the architecture-dependent 3A effect under each drug and its difference between drugs. Nuclear exposure, nascent RNA, TFE3 compensation, AMPK and Akt dependence, phosphosite identity and promoter occupancy are secondary or conditional modules with stated decision rules. Independent culture days are the unit of replication, and the primary estimands are prespecified genotype $\times$ architecture $\times$ stimulus contrasts.

Keywords: TFEB; EGFP; phosphorylation; experimental controls; nuclear localization; autophagic turnover

1 Rationale and Research Basis

TFEB is a member of the MiT/TFE transcription-factor family and a major transcriptional regulator of the autophagy–lysosome pathway. Under nutrient-rich conditions, mTORC1-dependent phosphorylation favors cytoplasmic retention of TFEB. During nutrient starvation or mTOR inhibition, TFEB accumulates in the nucleus and activates genes involved in lysosomal biogenesis and cellular clearance [36, 37]. Nutrient-dependent control of nuclear export also contributes to TFEB distribution [14].

HeLa cells expressing TFEB–GFP are an established system for measuring nuclear accumulation after starvation or Torin1 treatment [36]. This project uses that response to qualify a newly designed reporter, and the primary readout is the nuclear-to-cytoplasmic (N/C) distribution of the fusion protein. Additional modules measure reversibility, agreement with endogenous TFEB and downstream function. The construct is sequence-verified before any cell experiment.

1.1 Aims and Hypotheses

Aim 1 constructs and sequence-verifies an EGFP-tagged TFEB mammalian expression plasmid. **Aim 2** tests nutrient- and drug-dependent redistribution and recovery with reporter and endogenous measurements. **Aim 3** assesses the contribution of S142, S211 and the mTOR complexes, and the dependence of target-gene RNA and autophagic turnover on TFEB. **Aim 4** tests whether the effect of the C-terminal S466/S467/S469 cluster depends on the fusion architecture and on the stimulus, and attributes any architecture effect to the linker or to EGFP.

The replication hypothesis is that amino-acid starvation increases the N/C fluorescence ratio relative to time-matched nutrient-replete culture, that Torin1 increases it relative to matched vehicle, and that refeeding after starvation lowers it relative to continued starvation. The Aim 4 hypothesis is that substituting the C-terminal cluster changes target-gene RNA differently with and without the linker–EGFP fusion and differently under Torin1 and MK-8722. Aim 4 follows from unresolved interpretation of this cluster (Section 3); it has its own factorial design and decision rules.

2 Experimental Design

2.1 Overall Workflow

The study proceeds from reference-sequence retrieval and primer design to PCR amplification, plasmid construction, clone verification, transient transfection, and localization analysis. Sequence verification is the gate between molecular cloning and cell-based experiments.

2.2 Sequence-Defined Construct and Materials

The proposed construct, pEGFP-N1-TFEB-Bv1, expresses human TFEB followed by a C-terminal EGFP tag under the CMV promoter. The reference is TFEB transcript variant 1, NM_007162.3, whose CDS at transcript positions 303–1733 encodes NP_009093.1 (476 amino acids). A native, intron-free 1,431-nt CDS containing its original `TGA` is ordered as linear double-stranded DNA without codon optimization, for example through the gBlocks HiFi synthesis service. This defined template fixes the TFEB isoform at the outset. Supplier QC and the delivered sequence are checked on receipt [10, 15].

The empty backbone is pEGFP-N1, GenBank U55762.1 (4,733 bp). Kanamycin resistance selects in *E. coli*; neomycin/G418 resistance is used only for mammalian selection. The reference map includes the CMV promoter, EGFP, polyadenylation signals and replication origins [3, 9]. C-terminal EGFP has a precedent in TFEB localization experiments: Addgene #38119 (pEGFP-N1-TFEB) was built with HindIII/KpnI, whereas the present design uses HindIII/BamHI [7, 36].

The accepted *in silico* design is 6,134 bp (Figure 9). It retains the native TFEB start codon, removes the TFEB stop through reverse-primer placement, and connects TFEB to EGFP through the seven-residue `GDPPVAT` linker. EGFP's `ATG` remains an internal methionine codon, and its terminal `TAA` ends the 722-residue fusion. The forward primer adds `ACC` immediately before TFEB's start, giving an A at position -3 and G at $+4$. Reverse-primer `CC` contributes coding-strand `GG` to preserve the reading frame. Both junctions and the entire fusion were checked computationally.

The laboratory workflow (Figure 10) uses Q5 High-Fidelity $2\times$ Master Mix (NEB M0492) for cloning PCR, HindIII-HF R3104 and BamHI-HF R3136 in rCutSmart, Monarch cleanup and gel-extraction kits, T4 DNA Ligase M0202, NEB 5-alpha C2987 competent *E. coli* (a DH5α derivative),

Monarch T1110 minipreps and Sanger primer walking. OneTaq M0482 is used only for screening, and its products do not enter the construct.

2.3 Cloning Primers and High-Fidelity PCR

Table 1 gives the complete oligonucleotides and their functional parts. The forward annealing segment matches CDS positions 1–22; the reverse segment is the reverse complement of CDS positions 1407–1428, excluding the native stop. HindIII and BamHI have no internal sites in the selected TFEB coding segment and provide distinct 5′ cohesive ends. Six protective bases per primer exceed the three base pairs at which the manufacturer's end-cleavage table assigns both enzymes its highest category (50–100% cleavage under the stated oligonucleotide assay conditions) [17, 18, 22].

Table 1: Complete cloning primers and sequence-derived properties. Every sequence is written 5′ to 3′. In the partitioned lines, the order is protective bases, recognition site, deliberate addition, and template-annealing segment.

Primer	Sequence and properties
TFEB-HindIII-F	`ACTGACAAGCTTACCATGGCGTCACGCATAGGGTTGC`
	Parts: `ACTGAC` | `AAGCTT` | `ACC` | `ATGGCGTCACGCATAGGGTTGC`
	Full: 37 nt, 54.054% GC; annealing segment: 22 nt, 59.091% GC, $T_m = 69.498\,^{\circ}\mathrm{C}$. Full-oligo $T_m = 77.238\,^{\circ}\mathrm{C}$, not used for initial annealing.
TFEB-BamHI-R	`TATGACGGATCCCCCAGCACATCGCCCTCCTCCATG`
	Parts: `TATGAC` | `GGATCC` | `CC` | `CAGCACATCGCCCTCCTCCATG`
	Full: 36 nt, 61.111% GC; annealing segment: 22 nt, 63.636% GC, $T_m = 69.050\,^{\circ}\mathrm{C}$. Full-oligo $T_m = 78.896\,^{\circ}\mathrm{C}$, not used for initial annealing.

The T_m estimates use Primer3-py 2.4.0 with SantaLucia thermodynamics and salt correction, 50 mM monovalent ions, 2.0 mM Mg^{2+}, 0.8 mM total dNTP and 500 nM primer. The complete annealing segments and their terminal 12-mers each have one exact site in the defined template, and full-oligo hairpin and dimer screening passed the prespecified $55\,^{\circ}\mathrm{C}$ cutoff [34]. Only template-complementary segments guide initial PCR annealing, because the added tails do not bind the original template.

Reactions are assembled on ice using Table 2. The 1 ng template input targets the synthetic linear template, and the working dilution is calculated from the measured concentration. A no-template control replaces the template with water [31].

Table 2: Proposed Q5 cloning PCR, 50.0 μL total. Stock concentrations are targets and are adjusted to measured values.

Component	Volume	Final concentration or input
Q5 M0492 2× Master Mix	25.0 μL	1×; Mg^{2+} 2.0 mM; each dNTP 200 μM
TFEB-HindIII-F, 10 μM	2.5 μL	0.5 μM
TFEB-BamHI-R, 10 μM	2.5 μL	0.5 μM
Synthetic template, target 1 ng/μL	1.0 μL	1 ng total
Nuclease-free water	19.0 μL	To 50.0 μL

The program is 98 °C for 30 s; 30 cycles of 98 °C for 10 s, 65 °C for 20 s and 72 °C for 45 s; then 72 °C for 2 min. The 45 s extension rounds the expected 1,457-bp product upward at 30 s/kb. Annealing starts at 65 °C, with a 62/65/68 °C gradient if the product is weak or multiple [31].

2.4 Product Assessment, Restriction Digestion and Ligation

A 5 μL PCR aliquot is analyzed on 1.0% agarose in 1× TAE with 1× loading dye, a compatible DNA stain and 0.5 μg NEB N3232 1 kb ladder. Electrophoresis starts at 5 V/cm of electrode separation and stops after the relevant bands separate and before the tracking front exits. The ladder's 500/517 bp pair co-migrates. UV exposure of DNA intended for cloning is minimized [27].

A single 1,457-bp PCR product is purified with Monarch Spin PCR & DNA Cleanup T1130: 20–100 μL sample, five sample volumes of prepared BZ, an S1A column, two 200 μL WZ washes, and 20 μL EY elution after a 1 min wait. Centrifugation follows the current kit protocol at $16{,}000 \times g$ and room temperature. If additional bands occur, the target band is excised and purified with T1120 using four gel-slice volumes of BY, dissolution near 50 °C, an S1A column, two 200 μL WZ washes and 20 μL EY elution. Alcohol-containing buffers are prepared according to their current bottle labels. Recovered DNA is quantified and used at its measured concentration [23, 24].

PCR product and empty vector are digested separately with the common recipe in Table 3, at 37 °C for 60 min. Both enzymes have full listed activity in rCutSmart, and the enzyme volumes contribute 2% glycerol. Actual DNA and water volumes are recalculated from measured concentrations; insufficient material prompts a repeat or a scaled preparation. Uncut and single-digest analytical controls accompany troubleshooting of incomplete digestion. BamHI-HF cannot be heat-inactivated, so digestion products are purified before ligation [17, 20, 22].

Unpurified BamHI-HF digests for direct preparative or diagnostic gel loading receive SDS-containing Purple Gel Loading Dye B7024 (6×), diluted to 1× by adding one volume of dye to five volumes of sample; the SDS dissociates residual enzyme-bound DNA, and the no-SDS B7025 product is not used. Dye is added only to gel samples. If SYBR Safe or GelRed is used, staining follows electrophoresis as directed for B7024 [17, 21].

The 4,695-bp backbone is gel-purified to separate residual circular vector and the excised 38-nt MCS. The digested insert is purified with T1130. Its nominal retained length is 1,439 nt; the discarded top-strand terminal pieces are 7 and 11 nt. These pieces and the 38-nt MCS run below the resolving range of the gel. Both retained fragments have equal strand lengths; their pydna duplex spans are 4,699 and 1,443 bases because of staggered ends. Gel sizes use the nominal top-strand cut-to-cut lengths, and molarity uses mean strand lengths, which equal those nominal lengths for the selected ends. The distinct 5′ `AGCT` and `GATC` overhangs support directional ligation, so dephosphorylation is omitted; the vector-only ligation reports incomplete digestion or circular-vector carryover.

The initial ligation contains 20 fmol backbone and 60 fmol insert (3:1 insert:vector). With mean strand length L, the mass conversion is

$$m\ (\mathrm{ng}) = n\ (\mathrm{fmol}) \times L\ (\mathrm{nt}) \times 0.00066.$$

The calculated inputs are 61.974 ng backbone and 56.9844 ng insert. Table 3 gives a practical recipe rounded to 0.01 μL. The unrounded total DNA concentration is 5.948 ng/μL, within the manufacturer's 1–10 ng/μL starting range. Fully thawed ATP-containing ligase buffer is mixed before use. Ligation proceeds at 20–25 °C for 10 min, followed by 65 °C for 10 min to stop the reaction. A matched vector-only ligation replaces the insert volume with water. The restriction-cut ends carry 5′ phosphates, so no polynucleotide kinase step is needed [19].

Table 3: Proposed digestion and ligation mixtures. DNA stock concentrations are preparation targets adjusted to measured values.

Component	Volume	Amount or final condition
Each separate digestion: 50.0 μL		
DNA, target 50 ng/μL	10.0 μL	500 ng
10× rCutSmart	5.0 μL	1×
HindIII-HF, 20 U/μL	1.0 μL	20 U
BamHI-HF, 20 U/μL	1.0 μL	20 U
Water	33.0 μL	To 50.0 μL
Ligation: 20.00 μL		
Backbone, target 20 ng/μL	3.10 μL	Approximately 20 fmol
Insert, target 10 ng/μL	5.70 μL	Approximately 60 fmol
10× T4 DNA Ligase Buffer	2.00 μL	1×
T4 DNA Ligase M0202, 400 U/μL	1.00 μL	400 U
Water	8.20 μL	To 20.00 μL

2.5 Transformation, Screening and Plasmid Preparation

For each transformation, 2 μL ligation is added to 50 μL NEB 5-alpha C2987 cells. Cells remain on ice for 30 min, receive heat shock at $42\,^{\circ}$C for 30 s, return to ice for 5 min, and receive 950 μL SOC. Recovery proceeds at $37\,^{\circ}$C and 250 rpm for 60 min. Then 100 μL each of undiluted recovery and 1:10 and 1:100 SOC dilutions are plated on LB-kanamycin and incubated at $37\,^{\circ}$C overnight. Kanamycin is 30 μg/mL in bacterial plates and liquid cultures; G418 is reserved for mammalian cells [28, 29].

Vector-only ligation, a no-DNA control and the supplied pUC19 transformation control are processed in parallel. pUC19 is selected on ampicillin at 100 μg/mL. These controls report backbone background, contamination or selection failure, and competent-cell efficiency, respectively [28].

Eight isolated colonies are screened initially while numbered patches are retained. Each 25 μL OneTaq M0482 reaction receives trace colony material, 12.5 μL 2× mix, 0.5 μL of each 10 μM diagnostic primer and 11.5 μL water; final primer concentrations are 0.2 μM. Colony lysis is $94\,^{\circ}$C for 5 min, followed by 30 cycles of $94\,^{\circ}$C for 30 s, $58\,^{\circ}$C for 30 s and $68\,^{\circ}$C for 160 s, then $68\,^{\circ}$C for 5 min. The 160 s extension covers the longest modeled 2,608-bp product at 1 min/kb or longer. Annealing starts at $58\,^{\circ}$C, with a $55/58/61\,^{\circ}$C gradient if needed [30].

S-F1/S-R1 flank the insert, while S-F2/S-R1 pair an internal TFEB primer with a downstream vector primer to test orientation (sequences and sites in Table 4). Their model T_m values are $60.04/62.55\,^{\circ}$C and $66.09/62.55\,^{\circ}$C, respectively, using 50 mM monovalent ions, 1.8 mM Mg^{2+}, 0.8 mM total dNTP and 200 nM primer [34]. The correct construct predicts 2,605 bp for flanking PCR and 2,093 bp for orientation PCR; an empty vector predicts 1,204 bp for flanking PCR and no orientation product. Empty-vector and no-template controls accompany screening. In the first construction round, flanking-positive minipreps proceed to sequencing even if every orientation assay is negative; a sequence-accepted clone then serves as the orientation positive control in later rounds.

Up to three screening-positive colonies are grown independently in 3 mL LB-kanamycin at $37\,^{\circ}$C for 12–16 h with aeration; if fewer are recovered, those available are analyzed. For each T1110 miniprep, 1.5 mL culture is processed with 200 μL RNase-containing B1, 200 μL B2 for 1 min with gentle inversion,

then 400 μL B3 for 2 min. After clarification, DNA binds to an S2D column, receives 200 μL BZ and 400 μL WZ washes, and elutes in at least 30 μL EY. The manufacturer's additional 5 min BZ incubation is used for DNA intended for transfection. Yield and purity are measured [25, 26].

A 500-ng miniprep aliquot undergoes the defined 50-μL HindIII/BamHI digestion and gel analysis with the B7024 handling above. Agreement with 4,695- and 1,439-bp fragments confirms the insert and backbone sizes. Insert orientation, short indels, substitutions and the stop codon are resolved by sequencing, which is the acceptance gate.

Transfection-grade preparation. Each sequence-accepted clone is scaled from its glycerol stock to a midi-scale endotoxin-removal plasmid preparation for all cell experiments, with the kit and supplier fixed at procurement. Each lot receives A260/A280 and A260/A230 readings, a HindIII/BamHI diagnostic digest and Sanger reads across both junctions. Each construct has two independent lots, one from each accepted clone, which alternate across experimental days; lot numbers are recorded with every experiment so that no genotype is tied to a single preparation batch.

2.6 Sanger Primer Walking and Acceptance Criteria

Each candidate is sequenced with the ten primers in Table 4, in separate reactions. For Eurofins Standard Read purified-plasmid submissions, this construct falls in the 150–250 ng/μL stock range. A target of 200 ng/μL and 55 μL DNA per clone supports ten reads in the separate-template format ($10 + 9 \times 5$ μL). If needed, independently prepared aliquots from the same single-colony culture are pooled and measured or concentrated; different clones are never pooled. Each primer is supplied separately at 10 μM, with at least 10 μL for its first read and 5 μL per additional use. Current service requirements are rechecked before ordering [4].

Sanger-primer estimates use Primer3/SantaLucia with 50 mM monovalent ions, no added Mg^{2+} or dNTP, and 50 nM oligo. Selection required 18–25 nt, 35–60% GC, T_m of 50–62 °C, no four-base homopolymer, a terminal G/C without three consecutive terminal G/C bases, and modeled hairpin/homodimer $T_m < 40$ °C. Each oligo has one exact full-length binding site in the final construct [4, 34].

Table 4: Sanger and diagnostic primers. Coordinates are 1-based inclusive on pEGFP-N1-TFEB-Bv1. A plus/minus sign denotes the extension direction; every oligonucleotide is written 5′ to 3′. Windows are predicted usable read ranges.

ID	Complete sequence	nt; GC%; T_m (°C)	Binding site (strand)	Planned usable window
S-F1	GGGATTTCCAAGTCTCCACC	20; 55.00; 52.03	418–437 (+)	488–1087
S-F2	TGGGAACAAGTTTGCTGCCCAC	22; 54.55; 58.13	930–951 (+)	1002–1601
S-F3	ACCATCCTCAAGGCCTCTGTGG	22; 59.09; 57.96	1456–1477 (+)	1528–2127
S-F4	TGATCCACTTCTGTCCACCATGTCC	25; 52.00; 58.23	1971–1995 (+)	2046–2645
S-F5	GCTGGAGTACAACTACAACAGCCAC	25; 52.00; 57.83	2502–2526 (+)	2577–3176
S-R1	GACAAACCACAACTAGAATGCAGTG	25; 44.00; 54.94	2998–3022 (−)	2348–2947
S-R2	GTTGCCGTCCTCCTTGAAGTCG	22; 59.09; 57.87	2466–2487 (−)	1816–2415
S-R3	ATGGTGGACAGAAGTGGATCAGAGG	25; 52.00; 57.76	1967–1991 (−)	1317–1916
S-R4	ATCCACAGAGGCCTTGAGGATGG	23; 56.52; 57.86	1457–1479 (−)	807–1406
S-R5	TGATGTGGGCAGCAAACTTGTTCC	24; 50.00; 58.35	932–955 (−)	282–881

The planning window comprises extended bases 51–650 after each primer's 3′ end (600 usable bases), which excludes the early low-quality region and stays within a standard-read length. Adjacent

planned windows overlap by at least 50 nt. The acceptance target is positions 602–2799 (2,198 nt): 20 upstream vector bases, the 5′ junction and initiation context, and the full 2,169-nt fusion ORF through its final stop. The computational coverage check gives every target base at least one forward and one reverse read, with primer-binding bases covered by other primers. The outer S-F1 and S-R1 sites lie outside the target, and sequence beyond it derives from the unmodified reference vector [4, 6].

A clone is accepted when actual quality-trimmed chromatograms give unambiguous, concordant Q20-or-better calls in both directions across the entire target, no unexpected nucleotide changes, and the predicted restriction pattern. Reads are aligned to the exact B-v1 reference, with manual inspection of junctions and all mismatches [5]. Low-quality regions, single-strand gaps and ambiguous peaks trigger additional reads or primers. Reproducible mixed peaks prompt single-colony re-isolation and a fresh preparation. A clear unwanted coding or junction mutation rejects that clone; a shared error across independent clones prompts investigation of the template and design.

2.7 Troubleshooting

No cloning-PCR band prompts checks of template identity, dilution, primer matching and annealing conditions; additional bands prompt the gradient and target-band purification. High vector-only background prompts checks for incomplete digestion and residual circular DNA. Absence of colonies in both the test and positive transformation control points to handling or selection problems before a failed ligation is inferred. An apparently correct gel pattern with sequence disagreement fails acceptance. Only sequence-accepted plasmid enters the HeLa experiments.

2.8 Sequence-Defined Control and Mutant Constructs

Five plasmid conditions are defined: parental pEGFP-N1 expressing EGFP only, the accepted WT TFEB–EGFP reporter, and three derived reporters carrying S211A, S142A or S142A/S211A (Figures 1 and 2). Amino-acid positions refer to human NP_009093.1. The fusion constructs share the same 6,134-bp plasmid architecture, the GDPPVAT junction and C-terminal EGFP, and encode 722 residues. The single mutants and their combination are compared with WT under each treatment, which separates site-specific from combined contributions [2, 14].

Table 5: Designed codon changes; plasmid coordinates are 1-based inclusive.

Variant	Codon change	Plasmid nucleotides	Comparison enabled
S142A	AGC → GCC	1054–1056	Recovery with or without the S142 phospho-acceptor
S211A	TCC → GCC	1261–1263	S211-dependent distribution and 14-3-3 association
S142A/S211A	Both changes	Both intervals	Combined perturbation versus each single mutant

Each mutant is generated from a sequence-defined synthetic donor that carries the same cloning flanks as the WT amplicon, followed by the HindIII/BamHI workflow above. The donor FASTA files are in the design record. The WT cloning primers anneal outside the mutated codons and amplify each donor, so no mutagenesis primers are needed. Each batch includes a no-template PCR, a digested-vector-only ligation and the positive transformation control. Sanger reads cover the full fusion region and both junctions in both directions, with the target codons read explicitly. Two independently recovered, sequence-correct bacterial clones per genotype confirm the key contrasts and detect clone- or preparation-specific

effects; clone identity is randomized across experimental days. Bacterial clones serve as technical confirmation, and independent culture days remain the biological replicates.

The derived sequences keep the WT reading frame and the unique HindIII/BamHI cleavage geometry. Their digest and colony-PCR patterns therefore match WT, and full-length bidirectional sequencing identifies each mutant. Each construct passes three gates in order: sequence acceptance, intact full-length expression (anti-GFP and anti-TFEB blots), then phenotype. Alanine removes the serine phospho-acceptor and also alters local interactions; the RNAi and phospho-specificity modules address kinase identity. Phosphomimetic S-to-D/E variants are omitted because they do not reliably reproduce the phosphorylated state.

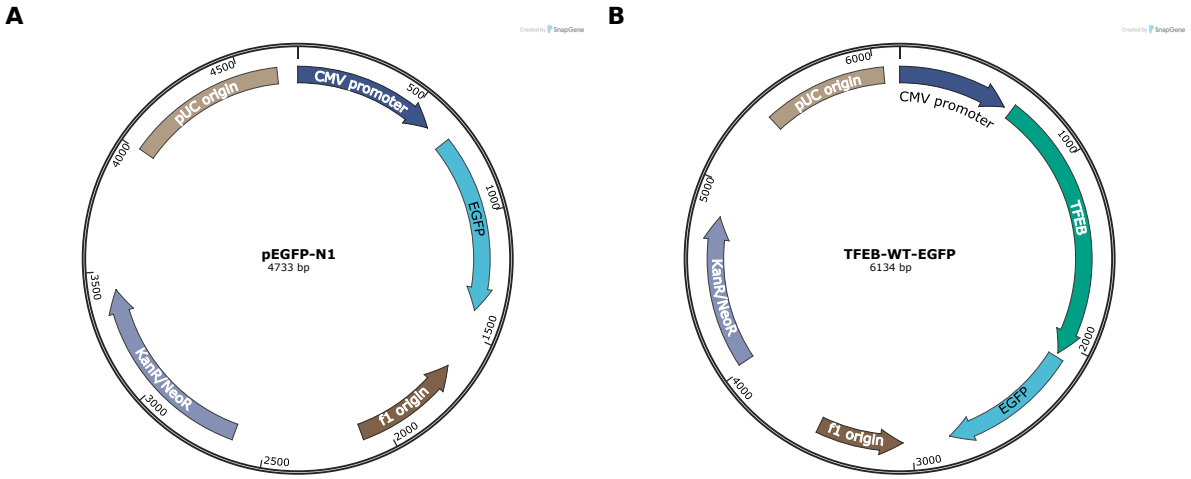


Figure 1: SnapGene Viewer maps of (A) the 4,733-bp EGFP-only control and (B) the 6,134-bp WT TFEB–EGFP reporter. Selected features keep their sequence coordinates and use a consistent NPG palette. The linker and minor regulatory annotations are omitted at this scale; complete GenBank records accompany the design.

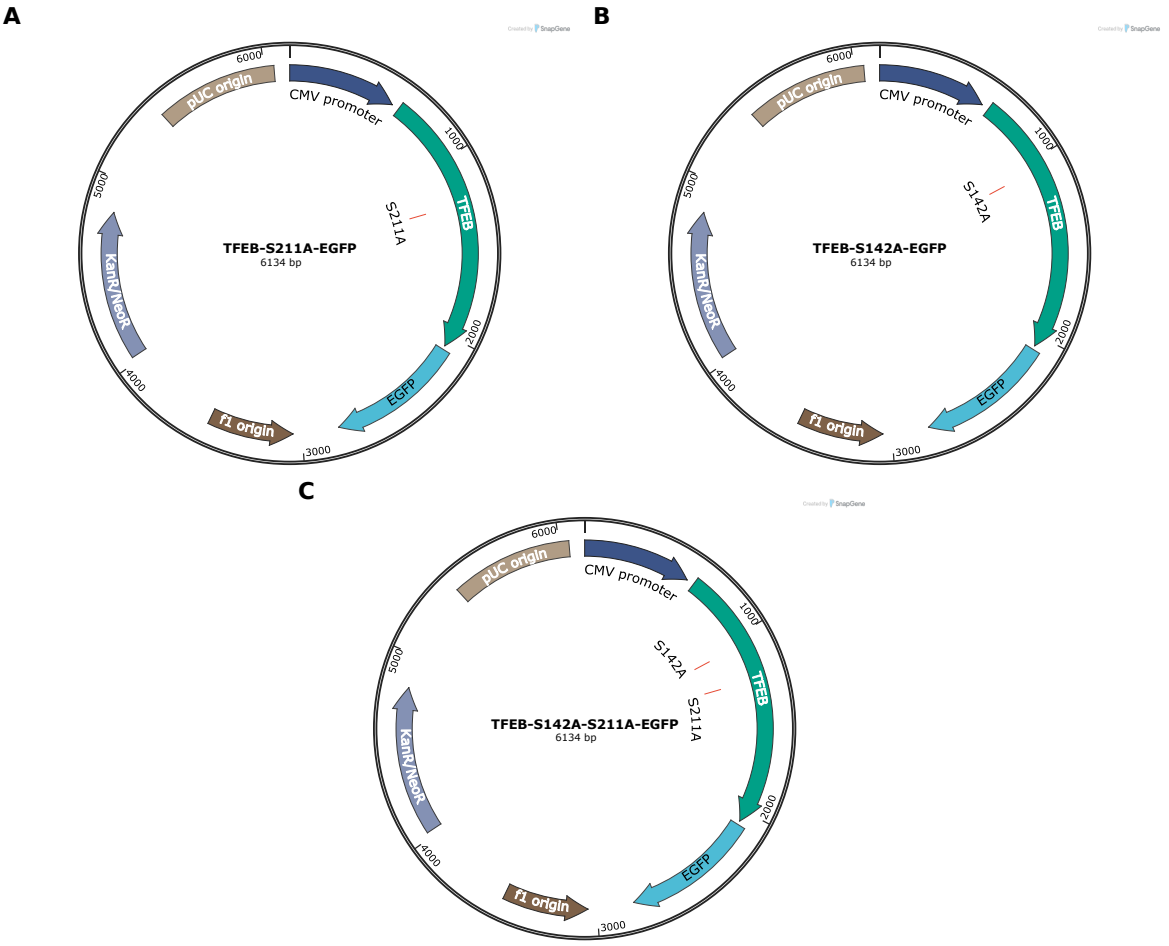


Figure 2: SnapGene Viewer maps of (A) S211A, (B) S142A and (C) S142A/S211A. Red marks identify the substituted codons; residue numbering refers to NP_009093.1. Every plasmid is 6,134 bp and encodes 722 residues; EGFP, backbone and linker sequences are identical to WT.

2.9 Questions and Control Structure

The expanded study tests reproducibility, reversibility, endogenous consistency, phosphorylation-site contributions, and TFEB-dependent function. Table 6 links each alternative explanation to a comparison. Modules use separate matched plates rather than crossing every factor in one experiment. Numerical cell-assay settings below are starting conditions; the pilot fixes them before confirmatory data are collected.

Table 6: Control matrix: comparison, readout and interpretation.

Question	Required comparison	Readout and decision
Background / reagent	Parental; reagent-only mock; EGFP-only; WT fusion, under corresponding treatments	Background, morphology and viability. Mock reports reagent effects; EGFP-only reports vector and fluorescent-protein effects.
Nutrient response	Amino-acid-replete vs depleted medium, with matched glucose, salts and dialyzed serum; separate serum-withdrawal arm	TFEB N/C and pathway markers. Serum withdrawal is analyzed as its own arm, separate from amino-acid deprivation.
Drug response	Torin1 vs equal DMSO; untreated vs DMSO	N/C, p-S6K/total S6K, viability. DMSO alone leaves N/C unchanged.
Reversibility	Starve 60 min, then refeed 15/30/60 min; continued starvation and continuous feeding at equal total times	Change from starvation-endpoint log N/C; recovery of signaling. Matched total times control for elapsed time and continued deprivation.
Tag / abundance	Endogenous TFEB in parental cells; reporter at two acceptable expression doses	Endogenous IF/fractionation; reporter expression-stratified N/C. Concordant reporter and endogenous responses validate the reporter.
Phosphorylation sites	WT, S211A, S142A and S142A/S211A, same backbone and treatments	Baseline, treatment interaction and recovery. Each mutant is compared with WT from its own starting N/C.
mTOR attribution	Non-targeting siRNA; two independent RPTOR siRNAs; two independent RICTOR siRNAs	Target depletion, p-S6K, p-AKT(S473), total counterparts and TFEB N/C. Both pathway perturbations are verified.
14-3-3 association	WT vs S211A, vehicle vs Torin1; EGFP-only; beads-only	GFP pull-down, input and recovered bait; bound 14-3-3 normalized to precipitated fusion.
TFEB dependence	Two individual TFEB 3′-UTR siRNAs; non-targeting control; empty vector; coding-only WT rescue; control-siRNA + WT	Endogenous depletion, rescue abundance and target transcription. Reproducible loss plus rescue supports a TFEB contribution.
Autophagic turnover	Each stimulus and knockdown/rescue condition with and without terminal bafilomycin A1	Inhibitor-dependent LC3-II increment, loading and viability. Flux is the increment with inhibitor.
Assay specificity	No-primary IF; phosphatase-treated lysate; S211A; fraction-purity markers; qPCR no-template/no-RT	Nonspecific staining, phospho-signal, mixed fractions and contamination are excluded before biological interpretation.

2.10 Cell Quality, Expression and Treatment

Use authenticated, mycoplasma-negative HeLa cells from a recorded source and restricted passage range. Distribute the same cell suspension across each experimental block. Culture in DMEM with 10% FBS at 37 °C and 5% CO_2, aiming for 60–70% confluence at transfection. Pilot 0.10, 0.25 and 0.50 μg fusion plasmid per 35-mm dish with one lipid reagent at its documented DNA ratio; evaluate expression after 24 h. The range follows published HeLa transfection work [36]. Select the lowest dose giving adequate nonsaturated signal, retain a second acceptable dose for sensitivity analysis, and compare the selected dose with endogenous TFEB abundance by anti-TFEB blot and IF; the reporter-to-endogenous ratio is reported. Plasmid molarity and total DNA mass are matched across constructs with a verified nonexpressing carrier, because equal mass gives different molecule numbers for the 4,733-bp empty and 6,134-bp fusion vectors.

The primary nutrient comparison uses a defined DMEM-based formulation with or without its amino-acid supplement, including glutamine; glucose, salts, vitamins and 10% dialyzed serum remain matched. Refeeding restores that amino-acid mixture after 60 min. A separate amino-acid-replete, serum-free arm tests serum withdrawal. The medium formulation and supplier record are frozen before execution. Fed and depleted wells receive matched washes and medium exchanges.

The Torin1 starting condition is 250 nM for 60 min, preceded by a 125/250/500 nM pilot at 30/60 min. Published work establishes Torin1 responsiveness of TFEB [2]; the pilot optimizes dose and duration for this transient-expression protocol. DMSO is matched to the highest concentration used, capped at 0.1%, in relevant controls. The lowest exposure with reproducible pathway engagement and acceptable viability is selected. Viability criteria are at least 90% membrane-intact cells and no more than a 10-percentage-point loss versus vehicle, including detached cells. Unsuitable conditions are rejected during the pilot, before confirmatory TFEB effects are inspected.

Acute localization is collected at 0/15/30/60 min in separate matched wells. For recovery, collect after 60-min starvation and after 15/30/60-min refeeding, with continuously fed and continued-starvation wells at 75/90/120 min total time. This series applies to WT and all three mutants. RNA, co-IP and flux use separate plates. Figure 3 shows acute and recovery sampling.

2.11 Imaging, Endogenous Confirmation and Sampling

Start with 4% paraformaldehyde fixation for 10 min and DAPI counterstaining; GFP preservation is checked in the pilot. Permeabilization and endogenous TFEB antibody dilution are optimized in parental cells. Include no-primary staining and show signal reduction with each TFEB siRNA before quantitative use. Acquisition settings are identical and nonsaturating within each day; plate positions and acquisition order are randomized, images are coded for analysis and raw files are retained. A 60-min leptomycin B arm on WT (starting at 20 nM, titrated in the pilot) defines the achievable nuclear ceiling, and each genotype's N/C is also reported as a fraction of that ceiling.

Nuclei are segmented with DAPI and cells with a compatible boundary marker or validated transmitted-light method, rather than a treatment-dependent GFP threshold. Nuclei and neighboring cells are excluded from cytoplasmic masks. N/C uses background-corrected mean intensities:

$$R_{\mathrm{N/C}} = \frac{I_{\mathrm{nucleus}} - I_{\mathrm{background}}}{I_{\mathrm{cytoplasm}} - I_{\mathrm{background}}}. \tag{1}$$

Component intensities are retained. Out-of-focus, saturated, mitotic, dead, overlapping or unsegmentable cells are excluded under pilot-frozen rules. Cytoplasmic intensity at or below background plus three

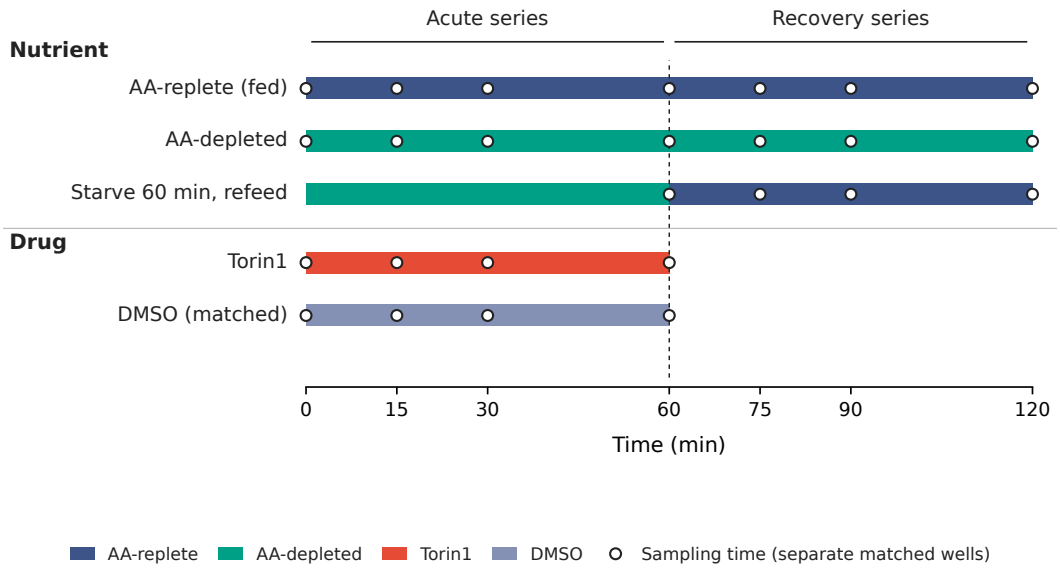


Figure 3: Planned sampling schedule. Bars show the exposure and open circles the sampling times, each from separate matched wells. The acute series (0–60 min) and the recovery series (60–120 min) include continuously fed and continued-starvation controls at matched total times. AA, amino acids; the dashed line marks the 60-min endpoint and refeeding. WT and all three mutants share the series, and transcription and flux assays use separate plates.

background standard deviations is nonquantifiable and its frequency is reported in place of an arbitrary large ratio. Exclusions are reported by group to expose treatment-dependent selection.

The plan is four independent culture/transfection days after a separate pilot, two technical wells per condition, and at least 30 eligible cells per well from at least five prespecified nonoverlapping fields. All eligible cells in those fields are analyzed, or a random subsample is drawn with a recorded seed. Cells, fields and technical wells are nested observations, and the independent culture day is the biological replicate. Key WT treatment/recovery contrasts are confirmed in parental cells by endogenous TFEB IF and fractionation; agreement of direction and size supports endogenous consistency.

2.12 Protein Expression, Phosphorylation and Fractionation

Anti-GFP and anti-TFEB whole-cell blots check intact expression. The WT fusion has 722 residues and a theoretical unmodified mass of 80.42 kDa; mutants have the same length and a slightly different exact mass. A low band is assigned as cleavage when anti-GFP and anti-TFEB blots identify complementary fragments. Linear detection is established with a pilot loading series, then loading starts at 15–20 μg protein per lane. Phosphorylation is preserved with appropriate inhibitors, uncropped membranes and markers are shown, and abundance is normalized to total-protein staining. A housekeeping band is used after its stability is checked.

p-TFEB(S211)/total TFEB is probed at the fusion-sized band and p-S6K(T389)/total S6K in the same lysates. S211A and a separate phosphatase-treated aliquot test phospho-antibody specificity; the phosphatase aliquot uses compatible buffer without inhibitor, and the untreated aliquot receives matched handling. The phospho/total ratio uses the fusion-sized band alone. p-S6K reports pathway engagement and p-TFEB(S211) reports the site.

Nuclear/cytoplasmic fractionation compares matched cell equivalents and corrects for fraction volumes and recovery. Lamin B1 or histone H3 marks nuclei; tubulin marks cytoplasm. Reciprocal contam-

ination is assessed and whole-cell input is retained. Redistribution is concluded from volume-corrected fractions with clean purity markers that agree with endogenous IF.

2.13 Site Perturbations, Genetic Attribution and Binding

The sequence-defined panel is described in Section 2.8. S211A tests loss of that phospho-acceptor; S142A and the double mutant test additional recovery-associated regulation. S211A raises basal nuclear localization, while further mTORC1 sites such as S122 keep TFEB under mTORC1 control [2]. Refeeding-related nuclear clearance measures net redistribution. S142/S138 have been linked to nuclear export [14]; fixed-cell N/C curves report net redistribution, and import and export rates are separated by live-cell cytoplasmic FRAP with unbleached/photobleaching controls or an export-inhibitor comparison.

For mTOR-complex attribution, two nonoverlapping siRNAs per target are tested individually against RPTOR or RICTOR, with equal-concentration non-targeting siRNA and reagent-only mock. The pilot starts at 10 nM for 72 h, with reporter transfection during the last 24 h. Depletion is verified in each experimental block, and p-S6K/total S6K and p-AKT(S473)/total AKT are assayed. Each depletion is compared with non-targeting control under fed conditions, with Torin1 as a response reference. A RICTOR result is interpreted when depletion is verified and a measurable mTORC2 readout is present; a matched stimulated positive-control lysate is included if basal p-AKT is too low. The genetic comparison is literature-supported [12]. RPTOR RNAi provides the mTORC1-specific test, since Torin1 inhibits both mTOR complexes. An RNAi-resistant RPTOR rescue construct, sequence-defined at procurement, gives the target-specific confirmation.

For co-IP, WT and S211A are compared under vehicle/Torin1 using GFP-affinity capture from equal total-protein inputs. EGFP-only, beads-only and input fractions are included; antibody-based IP adds a matched isotype control. 14-3-3 and recovered fusion are detected, and co-precipitated 14-3-3 is normalized to recovered TFEB–EGFP. Reduced association at equal bait recovery supports phosphorylation-dependent complex formation; bead-only binding and bait recovery are scored first.

2.14 TFEB Depletion, Rescue and Target-Gene Expression

Two independent siRNAs against the TFEB $3'$ UTR are tested individually, so the coding-only B-v1 transgene lacks the target sites. Before ordering, targets are verified against NM_007162.3 and expressed isoforms, off-target complementarity is checked and exact sequences are recorded. RNAi resistance requires both absence of the target site and maintained rescue expression during RNAi. siRNA sequences are verified at procurement.

For each of fed vehicle and sustained Torin1, the arms are: non-targeting siRNA + empty vector; each TFEB siRNA + empty vector; each TFEB siRNA + WT; and non-targeting siRNA + WT. Reagent-only controls are included in the pilot. Plasmid molarity and total DNA are matched; endogenous depletion is quantified separately from fusion expression, and rescue is titrated close to endogenous abundance. The control-siRNA + WT arm detects effects of rescue overexpression. RNAi runs 72 h with WT rescue during the final 24 h. Mutant rescue follows successful WT rescue.

MCOLN1, *LAMP1* and *CTSD* transcripts are measured at 4/8 h on separate plates with matched time controls. The panel and timing test the output of the established TFEB transcriptional program [38]. *RPLP0*, *HPRT1* and *TBP* are piloted, and at least two stable reference genes are retained. New qPCR primers are validated for specificity, a single product, 90–110% efficiency and quantitative range, with clean no-template/no-RT controls, documented RNA quality and an inter-run calibrator [1]. Exact oligos are fixed at procurement. Normalized Cq differences are analyzed, with $2^{-\Delta\Delta Cq}$ used when efficiency

assumptions hold. Biological n is the independent culture day; technical PCR wells are averaged. These mature-RNA measurements integrate synthesis and decay; the nascent-RNA controls of the research extension address transcription rate.

The pilot also compares TFEB-only with combined TFEB + TFE3 depletion for residual drug-induced MCOLN1. If TFEB-only depletion retains at least half of the induction (pilot estimate), the combined background is adopted and frozen before confirmation. A response reduced by both TFEB siRNAs and restored by WT supports a TFEB contribution. Residual induction is attributed to incomplete depletion, TFE3/MITF compensation or parallel regulation, and TFE3 abundance/localization are assayed when substantial residual induction persists. A failed rescue is interpreted after rescue abundance, localization and RNAi resistance are confirmed.

2.15 Autophagic Turnover and Toxicity

On separate plates, LC3-I/LC3-II and SQSTM1/p62 are measured with and without a terminal 60-min pulse of 100 nM bafilomycin A1 during a 4-h Torin1 exposure, following consensus autophagy-flux practice [11]. A first pilot tests 50/100 nM for 30/60 min for blockade and tolerability. Each stimulus and knockdown/rescue condition has its own inhibitor-free and inhibitor-treated well: vehicle, bafilomycin alone, stimulus alone and combination. Operational turnover is the within-condition inhibitor-dependent increment of normalized LC3-II, compared across stimulus and TFEB loss/rescue. This paired-inhibitor logic follows experimental TFEB/autophagy work [38].

p62 transcript is measured alongside protein, since transcription changes alter protein abundance. Viability and loading are checked in every arm. More LC3-II without additional inhibitor-dependent accumulation points to impaired clearance. mTOR inhibition also acts through TFEB-independent pathways, so TFEB depletion may leave part of the response intact. Localization is scored in bafilomycin-free wells. If turnover remains ambiguous, a tandem mCherry–GFP–LC3 assay with inhibitor and single-channel controls provides an orthogonal readout (reporter plasmid selected at procurement).

2.16 Prespecified Analysis and Decision Rules

Primary contrasts are WT amino-acid starvation vs replete medium at 60 min; WT Torin1 vs vehicle at 60 min; and WT 30-min recovery vs continued starvation at 90 min total time. Cell $\log_2$(N/C) is averaged within well and then across independent day/condition. Paired day-level contrasts are analyzed with effects and 95% confidence intervals, and Holm correction applies across the three primary tests. Four independent days is the minimum; pilot variance and the effect threshold below set the confirmatory sample size before that dataset is collected.

Success criteria for the replication. The reporter is accepted as responsive when each primary contrast has a Holm-adjusted $p < 0.05$, a mean day-level change in $\log_2$(N/C) of at least $\log_2 1.5$ in the expected direction (increase for starvation and Torin1, decrease for recovery), and the same sign on every independent day. The pilot may revise the $\log_2 1.5$ threshold, and any revision is recorded before confirmation. Reporter acceptance also requires matched direction in endogenous TFEB IF and clean fractionation.

For mutants, a genotype-by-condition/time model uses day as a block and prespecified WT-versus-mutant interaction contrasts. Recovery analysis accounts for each genotype's starvation-endpoint ratio, since different initial ratios do not by themselves indicate different recovery rates. If sample size or model assumptions fail, day-level contrasts with uncertainty are reported and the planned extra replicates are collected. RNAi/rescue, transcripts, co-IP and turnover are separate secondary families with multiplicity

control. Plots show independent-day summaries with nested cell distributions. Equivalence claims use a prespecified margin. Discordant assays trigger the relevant control checks.

2.17 Execution Plan and Deliverables

The core program is organized into twelve modules with explicit outputs (Table 7). The eight-week timeline starts once reagents are available. Six additional research modules are specified in the next section. Modules advance after their stated acceptance checks. Figure 4 connects the levels of evidence; arrows indicate the planned order of investigation.

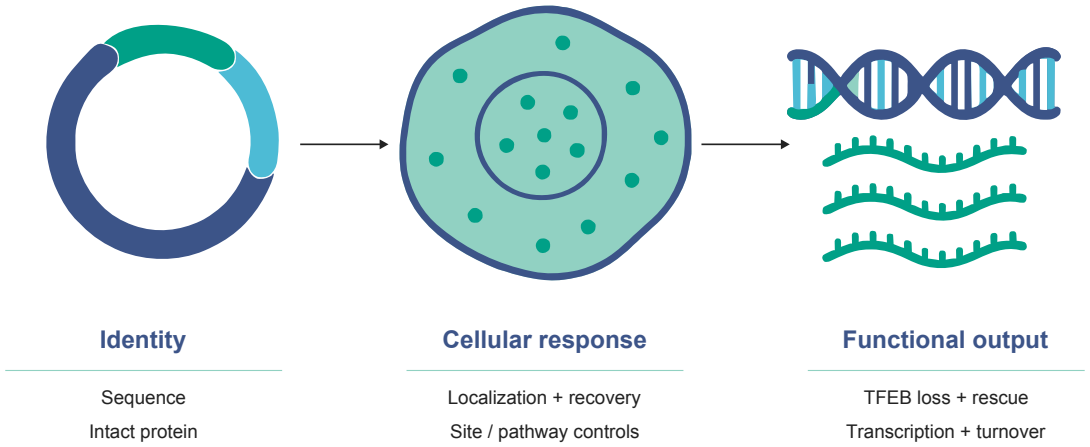


Figure 4: Evidence structure for the expanded study. Construct identity precedes biological assays; reporter and endogenous localization are checked before site/pathway interpretation. TFEB loss/rescue tests target-gene RNA responses and turnover. The transcription branch requires qualified nascent-RNA evidence for a rate interpretation, and turnover requires paired bafilomycin A1 controls. Icons are conceptual drawings. Each stage uses the negative, positive and specificity controls in the methods; agreement strengthens interpretation and disagreement narrows the conclusion.

Table 7: Proposed workload, acceptance checks and assessable deliverables.

ID	Week	Module	Deliverable / acceptance check
M01	1	WT and three mutant donor/construct designs	Versioned sequences, five plasmid maps, codon and junction audit; material identity verified on receipt.
M02	1–2	Cloning, selection and clone QC	PCR/digest records, failed-clone log and full bidirectional sequence acceptance; two independent accepted clones per genotype for key checks.
M03	2–3	Cell quality and expression pilot	Authentication/mycoplasma record; dose, viability and detector-linearity decisions frozen before confirmation.
M04	3–4	WT nutrient and drug response	Matched-media/solvent controls; raw images, masks, per-cell intensities and independent-day N/C summaries.

ID	Week	Module	Deliverable / acceptance check
M05	4	Endogenous TFEB and fractionation	Specific endogenous IF; reciprocal fraction-purity checks and volume-corrected partition estimates.
M06	4–5	Time course, refeeding and mutants	Four-genotype baseline/treatment and recovery contrasts; matched elapsed-time controls and clone-confirmation record.
M07	5	Phosphorylation and pathway readouts	Uncropped p-TFEB, p-S6K and total-protein blots; S211A and phosphatase specificity checks.
M08	5–6	RPTOR versus RICTOR RNAi	Two individual siRNAs per target, knockdown and pathway-engagement evidence; complex-attribution conclusion.
M09	6	14-3-3 association	GFP capture with input, EGFP-only and beads-only controls; association normalized to recovered bait.
M10	6–7	TFEB loss and WT rescue	Two independent 3′-UTR siRNAs; endogenous depletion, verified rescue expression and control-siRNA + WT arm.
M11	7–8	Transcription and turnover	Validated qPCR assays and paired inhibitor/no-inhibitor LC3-II contrasts; p62 transcript/protein and viability.
M12	8	Integration and reproducibility	Blinded analysis record, exclusions, effect estimates with intervals, source data, figure legends and alternative-interpretation table.

As a workload example, the four fusion genotypes crossed with three matched 60-min conditions (fed vehicle, amino-acid deprivation with vehicle, and fed Torin1) yield 12 conditions. Four independent days with two technical wells per condition require 96 assay wells and at least 2,880 eligible cells at 30 cells per well. These quantities exclude parental/mock/EGFP-only controls, pilot wells, recovery time points, protein/RNA plates and repeat attempts; a separate plate map enumerates those additions before execution. Independent days supply the replication, and additional cells from one day add precision within that day only. Splitting the larger program into modules keeps each comparison interpretable and the schedule feasible.

The report supplies design figures and computational sequence checks. Experimental image, blot, qPCR and turnover panels are planned outputs filled with acquired data. Before M08–M11, exact siRNA targets, qPCR oligos and reagent identities are finalized and their performance verified.

3 Hypothesis-Driven Research Extension

3.1 From Replication to a Discriminating Question

The localization experiments above reproduce established TFEB biology. The extension asks a different question: *does the effect of the C-terminal S466/S467/S469 cluster on target-gene RNA abundance depend on the stimulus and fusion architecture?* Protein abundance and nuclear exposure are measured as possible mediators, and the architecture effect is the hypothesis under test.

Paquette et al. reported that AMPK-dependent phosphorylation of S466/S467/S469 governs TFEB transcriptional activity, whereas mTORC1 governs cytosolic retention; a GFP-tagged S466A/S467A/S469A mutant (3A) translocated to the nucleus yet lost transcriptional activity [16, 33]. Negoita et al. subsequently found that AMPK activation dephosphorylates this cluster, that a synthetic peptide containing the sites is a poor AMPK substrate in vitro, and that AMPK loss abrogated the MK-8722 response but not the Torin1 response [16]. These findings come from different cell systems, stimuli and constructs, and the matrix below supplies a matched comparison. 3A is an existing mutant. S467 is also an Akt phosphorylation site that restrains TFEB nuclear entry independently of mTORC1 [32], so stimulus-dependent 3A effects can reflect Akt as well as AMPK; p-AKT(S473) is therefore measured under both drugs.

The proposed contribution is a controlled genotype–architecture–stimulus comparison in HeLa cells, followed by selected mechanistic tests. Negoita et al. also obtained supporting phosphosite evidence from C-terminal GFP knock-in material [16], so GFP is treated as one of several candidate sources of the literature disagreement. The 3T substitution retains hydroxyl-bearing side chains and remains phosphorylatable, which makes it a sequence/chemistry probe; Negoita et al. used S466T/S467T and S469T substitutions as phospho-antibody specificity controls in HEK293 cells, and observed that S469A raised the p-S466/S467 signal [16]. A targeted literature search identified no matched comparison of these three factors.

The two literature models predict opposite outcomes for 3A. If AMPK-dependent phosphorylation activates TFEB, as Paquette et al. proposed, 3A loses target-RNA induction under both drugs. If dephosphorylation activates TFEB, as Negoita et al. proposed, a non-phosphorylatable 3A retains or exceeds WT induction. A fusion-architecture effect appears as a different sign or size of the 3A–WT difference between tagged and untagged TFEB. The primary contrasts below are two-sided, and these predictions guide their interpretation.

The two architectures differ in linker, EGFP coding sequence, protein length and the position of the stop junction, so "tag" in the matrix denotes the entire GDPPVAT–EGFP fusion architecture. Attributing an effect to EGFP itself is the purpose of module R06, which adds two linker-only constructs once an architecture effect is established. Drug-associated changes in transgene RNA or protein stability are checked before a protein-level tag mechanism is proposed.

3.2 Five Additional Sequence-Defined Constructs

The existing WT fusion is reused. Five additional plasmid designs provide the remaining cells of the six-construct matrix (Table 8). All retain the same CMV promoter and bacterial backbone. Full GenBank, DNA FASTA and translated protein files are supplied in `research-v3`; circular maps are shown in Figures 5 and 6.

The three codons occupy bp 2026–2028, 2029–2031 and 2035–2037; intervening F468 is unchanged. Tag removal shortens the plasmid by 738 bp and eliminates the original BamHI site, so the WT HindIII/BamHI diagnostic and its fragment sizes do not apply to the untagged constructs. The procurement route for this extension is sequence-defined whole-plasmid synthesis followed by independent whole-plasmid sequence verification of every altered codon, the new stop junction and the backbone. Two independently recovered, sequence-accepted bacterial clones per construct confirm key comparisons as technical replicates; independent culture days remain the biological replicates.

Table 8: Research constructs and exact intended changes. Coordinates refer to the accepted 6,134-bp WT fusion.

Construct	bp / aa	Intended difference from the WT fusion
3A–EGFP	6134 / 722	S466A/S467A/S469A; each AGC codon becomes GCC.
3T–EGFP	6134 / 722	S466T/S467T/S469T; each AGC codon becomes ACC.
WT, untagged	5396 / 476	Replace bp 2059–2799 (linker plus EGFP including stop) with the native TFEB stop codon TGA.
3A, untagged	5396 / 476	The same tag removal plus the three AGC-to-GCC changes.
3T, untagged	5396 / 476	The same tag removal plus the three AGC-to-ACC changes.

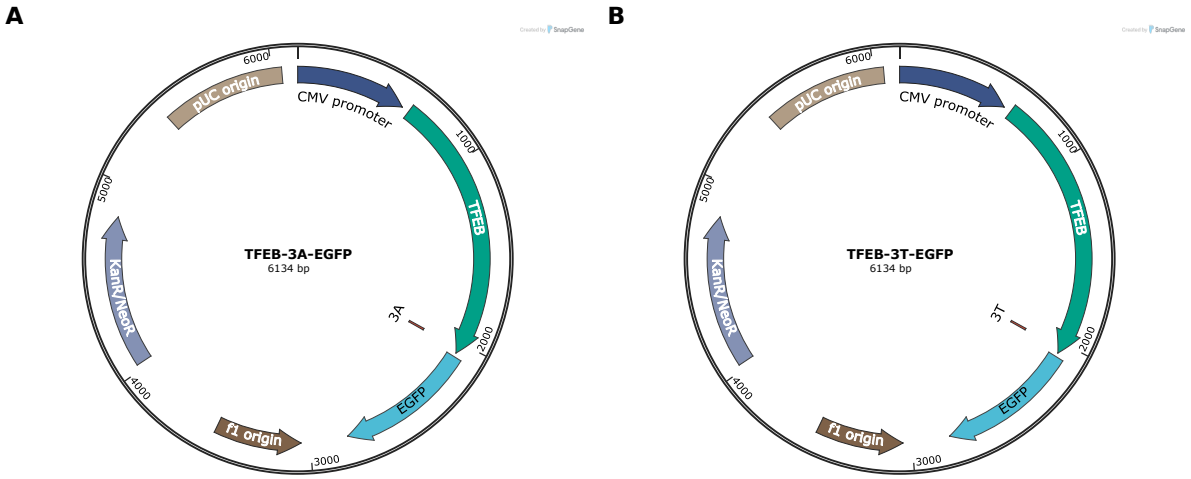


Figure 5: Additional tagged constructs rendered by SnapGene Viewer. (A) TFEB-3A–EGFP; (B) TFEB-3T–EGFP. Both are 6,134 bp. The red feature marks the S466/S467/S469 cluster; its span includes unchanged intervening sequence. CMV is blue, TFEB teal, EGFP cyan, selection purple and replication origins brown.

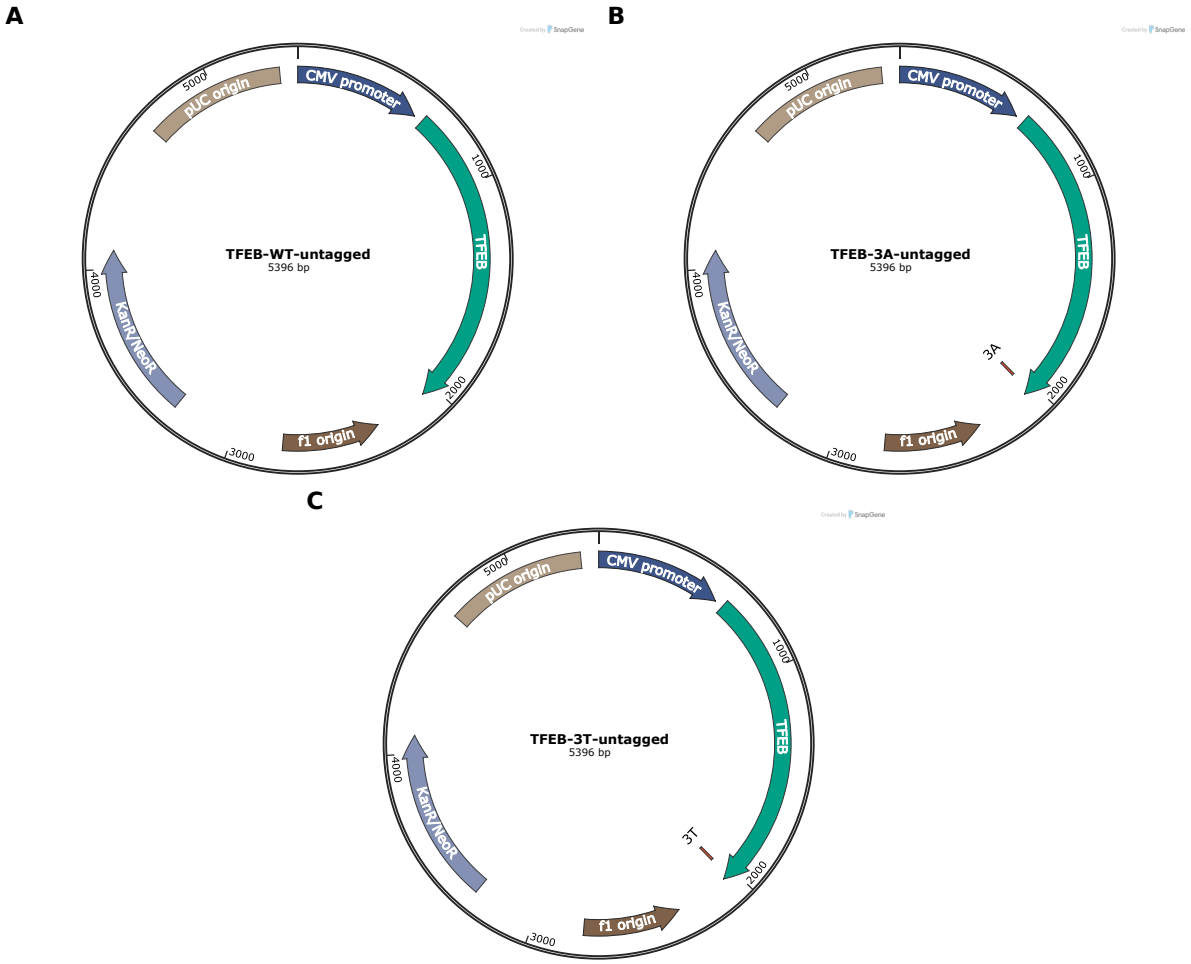


Figure 6: Matched untagged constructs rendered by SnapGene Viewer: (A) WT, (B) 3A and (C) 3T. Each is 5,396 bp and encodes a 476-residue TFEB protein followed by its native stop codon. Removing the linker and EGFP preserves the remainder of the parental backbone while changing downstream coordinates and the restriction pattern. Feature colors match the tagged maps. Full annotations and independent sequence checks accompany the design files.

3.3 Core Factorial Comparison and Controls

WT/3A/3T TFEB with EGFP-tagged/untagged architecture and vehicle/Torin1/MK-8722 treatment give 18 core conditions (Figure 7). The principal comparison is performed after endogenous TFEB depletion with one validated 3′-UTR siRNA that has no site in the coding-only constructs. The complete prespecified WT/3A comparison (both architectures and all three treatments) is repeated with a second individual TFEB siRNA in separate culture-day blocks. Transcript coverage across expressed TFEB isoforms and maintained transgene expression during RNAi are verified. Endogenous depletion is quantified by 3′-UTR RT-qPCR and by protein in matched no-rescue wells. Untagged rescue and residual endogenous TFEB co-migrate, and the no-rescue wells bound the residual endogenous level.

Acceptance gates are acceptable viability, at least 80% endogenous protein depletion in no-rescue wells and a pretreatment median TFEB signal per expressing cell within 0.5–2 times the parental reference. Untagged WT must restore at least half of the pilot-estimated loss of drug-induced MCOLN1 response, in the same direction, in each drug context for which a TFEB-dependent interpretation is sought, and the loss must exceed the pilot assay noise. If this is not met for a drug, the comparison for that drug remains descriptive until the assay is repaired. If MK-8722 fails the gate, the extension proceeds with Torin1 alone, the primary estimands reduce to the architecture-dependent 3A effect under Torin1 (defined below), and the AMPK module is dropped. A tagged WT that rescues poorly is retained as an architecture result. The knockdown background (TFEB-only or TFEB + TFE3, chosen in the core pilot) and all gates are frozen before confirmatory genotype outcomes are unblinded.

The following controls are run at vehicle, Torin1 and MK-8722 in each independent block. A machine-readable condition manifest accompanies the proposal; counts include culture conditions, and qPCR technical reactions are counted separately.

Table 9: Research condition inventory. NT is non-targeting siRNA; KD is the selected individual TFEB siRNA (with TFE3 siRNA added if the pilot adopts combined depletion). A validated nonexpressing DNA carrier supplies the no-rescue DNA control.

Arm, crossed with three treatments	Primary	Second-siRNA confirmation
KD + WT/3A/3T, both architectures	18	12 (WT/3A)
NT + WT/3A, both architectures	12	12
NT or KD + nonexpressing carrier, no rescue	6	6
NT or KD + EGFP-only	6	6
Reagent-only mock, neither nucleic acid	3	3
Total conditions per independent day	45	39

NT + WT/3A arms test whether a genotype effect depends on the depletion background; no-rescue arms test loss and baseline drug responsiveness. EGFP-only and the nonexpressing carrier are separate controls: the first reports fluorescent-protein effects and the second reports TFEB loss. An untreated/no-solvent comparison is added in the pilot. DNA carrier identity, total DNA mass and total siRNA concentration are constant within the DNA-transfected comparisons, and reagent-only mocks lack nucleic acid by design. DNA molarity is the starting calibration, and expression titration then sets the working dose. Positions and processing order are randomized, every contrast is balanced across plates, and shared reference samples are included if a complete day block spans plates. Plasmid lot, endotoxin/purity QC and clone identity are recorded; each genotype uses lots from both accepted clones, alternated across

days. Key effects are confirmed with the second sequence-accepted clone.

Pilot DNA amounts are set separately to overlap TFEB abundance and the fraction of expressing cells across constructs, using matched total DNA and transfection reagent. Both architectures are quantified with the same anti-TFEB reagent, recognizing an epitope outside the changed C-terminal region, with specificity validated by depletion and linearity by antigen-loading series. Each construct-specific DNA dose is frozen before drug exposure and used in all three treatment arms; retitration after treatment would erase a biological abundance response. In the pretreatment expression pilot, the target is a between-architecture ratio of 0.8–1.25 for the median TFEB signal among expressing cells and an expressing-fraction difference no greater than 10 percentage points. The measured differences and their uncertainty are reported. A common threshold above depleted/no-rescue background is frozen, fields are selected randomly, and all-cell distributions, stratified summaries and exclusions are reported. GFP brightness is not used to select one architecture. Bulk RNA from transiently transfected wells averages over transfected and untransfected cells, so the transfected fraction is measured for every construct. If common expression support is absent, delivery is revised before the confirmatory comparison, and the constructs move to single-copy isogenic lines (Flp-In-type integration) or low-MOI lentiviral pools, which deliver matched dose without transient-transfection variance. At baseline and 6 h, transgene RNA is assayed with validated backbone–TFEB-junction and shared TFEB-CDS assays, endogenous RNA with the 3′-UTR assay, and protein separately; tag-dependent RNA abundance is the alternative to altered protein activity.

For the extension, a common nutrient-replete medium and matched final solvent are used. MK-8722 is piloted at 1, 3 and 10 μM, and the Torin1 125/250/500-nM pilot range is retained. The lowest tolerated dose with reproducible intended engagement is selected; these doses are HeLa starting points fixed by the pilot. p-ACC/total ACC and p-S6K/total S6K are measured under both drugs, together with p-AKT(S473)/total AKT, p-TFEB(S211), TFEB abundance, cell counts and viability. Engagement and tolerability are rechecked at 6 h, since an acute pilot qualifies only the acute exposure. The two drugs are compared at their own pilot-selected doses, with dose and kinetic differences reported and a second acceptable dose as the sensitivity experiment. Torin1 also inhibits mTORC2/Akt signaling, so the drug comparison tests stimulus dependence, and pathway attribution comes from the AMPK and Akt modules below.

Localization is monitored at 0, 30, 60, 120, 240 and 360 min in matched destructive samples, and the primary RNA endpoint is collected at 6 h. N/C, nuclear concentration, integrated nuclear TFEB signal and cell/nuclear size are retained, because N/C can be equal when nuclear amounts differ. The sampled nuclear-signal trajectory and its trapezoidal area summarize population exposure; they are a population proxy, whereas single-cell trajectories and absolute molecule counts require live-cell imaging and calibration. An intrinsic transactivation interpretation requires overlapping nuclear-signal distributions and comparable trajectories, supported by orthogonal fractionation. Matching limits are prespecified from assay precision before confirmation. Exposure differences are reported as a mediator, and all samples stay in the primary analysis.

3.4 Primary Estimand, Replication and RNA Interpretation

The primary endpoint is normalized steady-state RNA abundance of the primary target at 6 h. MCOLN1 is the default; LAMP1 and CTSD are secondary. The pilot confirms MCOLN1 or replaces it once, with the target among MCOLN1, LAMP1 and CTSD that shows the largest TFEB-dependent drug induction in non-targeting, WT-rescue and KD-no-rescue wells, using no 3A or 3T data. The selec-



Figure 7: Factorial design for the research extension. (A) WT, 3A and 3T TFEB drawn to scale with the EGFP fusion, untagged or linker-only (R06, conditional) architecture: TFEB (teal), GDPPVAT linker (coral), EGFP (cyan), stop codon (black) and the S466/S467/S469 cluster (orange tick). (B) Matched conditions: filled circles mark the 18 core conditions and dashed circles the conditional R06 arms; Table 9 adds the full depletion/rescue, background and mock controls. (C) Weights of the primary contrasts on the WT and 3A rows: A_M and A_T are the architecture-dependent 3A effects under MK-8722 and Torin1, and $C = A_M - A_T$ takes the A_M weights on the MK-8722 cells and the negated A_T weights on the Torin1 cells (filled $+$, $+1$; open $-$, -1). 3T rows are exploratory. Expression, pathway and assay checks are specified in the text.

tion is frozen before confirmation, and if none of the three shows induction above pilot noise, R02 does not proceed. RNA integrity, genomic-DNA removal, primer efficiency, quantification limits and at least two reference genes are validated across all treatments, architectures and depletion backgrounds, following qPCR reporting principles [1]. No-template/no-RT controls and an inter-run calibrator are included. A missing amplification is recorded as missing, and values outside the validated quantitative range are flagged under a prespecified repeat/exclusion rule.

Let $Y_{g,a,s,d}$ denote the mean $\log_2$ normalized RNA value from culture wells on independent day d, for genotype g, architecture a (E, EGFP fusion; U, untagged) and stimulus s (V, vehicle; M, MK-8722; T, Torin1). qPCR technical replicates are averaged within each sample first. Define

$$\Delta_{a,s,d} = (Y_{3A,a,s,d} - Y_{3A,a,V,d}) - (Y_{WT,a,s,d} - Y_{WT,a,V,d}), \tag{2}$$

$$C_d = (\Delta_{E,M,d} - \Delta_{U,M,d}) - (\Delta_{E,T,d} - \Delta_{U,T,d}). \tag{3}$$

The architecture-dependent 3A effect at stimulus s is

$$A_{s,d} = (Y_{3A,E,s,d} - Y_{WT,E,s,d}) - (Y_{3A,U,s,d} - Y_{WT,U,s,d}), \tag{4}$$

and $C_d = A_{M,d} - A_{T,d}$. The three primary estimands are the means of $A_{M,d}$, $A_{T,d}$ and C_d: whether the 3A effect differs between architectures under each drug, and whether that architecture dependence

differs between the drugs. Holm correction applies across these three tests. $A_{s,d}$ uses stimulus-matched cell means only, and vehicle terms cancel algebraically from C_d and remain essential for estimating each drug response and the baseline genotype effects. The component effects are reported alongside C_d, because equal nonzero architecture dependence under both drugs gives $C_d = 0$. The primary result is the total effect in the calibrated expression system. Post-treatment nuclear exposure and protein abundance are mediators and are not entered as covariates in the primary model; matched-exposure analyses are secondary.

Complete independent-day blocks use separate cell culture, RNAi and transfection preparations, with technical wells distributed across plate positions. Day-level C_d values, their mean and a two-sided 95% interval based on between-day variation are reported (a t interval with $n - 1$ degrees of freedom when its assumptions hold). Biological n is the independent day; wells, cells, clones and PCR reactions are averaged within day. A failed well is repeated, and cell-level data do not replace it. Repeat rules are prespecified from technical QC; if a required arm is unavailable, that incomplete day is excluded from the paired primary estimate, kept descriptively with the reason recorded, and a replacement complete block is acquired. Outcome size or direction is never an exclusion criterion.

A separate pilot of at least four complete day blocks estimates the variance of C_d, and the confirmatory sample size is fixed before confirmation data are collected. $A_{s,d}$ combines four cell means and C_d combines eight, so independent errors of variance σ^2 per mean give $\mathrm{Var}(A_{s,d}) = 4\sigma^2$ and $\mathrm{Var}(C_d) = 8\sigma^2$; C_d therefore sets the sample size. The number of days n for a 95% interval half-width $h = 0.5 \log_2$ units satisfies $n \geq (t_{n-1,0.975}\, s_C/h)^2$, where s_C is a conservative pilot estimate of the standard deviation of C_d. Six independent days is the minimum, and n increases as the calculation requires. The precision target is fixed in advance, the calculation is retained, and data collection continues to the fixed sample size regardless of significance. If resources cannot meet the target, the extension is reported as exploratory. Pilot and confirmatory data are kept separate. The secondary family comprises the four MCOLN1 $\Delta_{a,s}$ contrasts and the primary-form A_M, A_T and C contrasts for LAMP1 and CTSD (ten tests, Holm adjustment). 3T and non-targeting-background comparisons are exploratory. The second-siRNA confirmation re-estimates the same primary contrast with its own complete blocks and interval; the two siRNAs are reported side by side and not pooled.

Synthesis versus stability. Mature RNA integrates production and degradation [35]. Before a difference is called a transcriptional defect, the prespecified WT/3A contrasts are repeated with a short, pilot-qualified 4-thiouridine pulse ending at 6 h, and newly synthesized MCOLN1 and a secondary target are quantified. Controls are matched unlabeled/background-purification samples, input RNA, labeled/unlabeled recovery standards, cell-count normalization, and checks for labeling toxicity and treatment-dependent incorporation (recovery standards control purification, and incorporation is compared across treatments to control cellular nucleoside uptake). Pulse duration and performance criteria are frozen before confirmation; if incorporation differs between treatments, the assay does not support differential synthesis. Concordant nascent and mature RNA supports a production interpretation, and discordance triggers RNA-stability/processing studies. Intronic RT-qPCR is an exploratory precursor-RNA assay. Without a qualified nascent-RNA assay, the endpoint is reported as target-gene RNA abundance.

MiT/TFE compensation. Overexpressed TFE3 induces MCOLN1 and other lysosomal genes, including in TFEB-depleted HeLa cells [13]. TFE3 RNA, protein and localization are measured in the prespecified WT/3A comparison and the no-rescue controls. The knockdown background (TFEB-only or TFEB + TFE3) is chosen in the core pilot and frozen. When TFE3 activity or residual induction changes between genotypes, the key contrasts are repeated with TFEB-only, TFE3-only and combined depletion at matched total siRNA, two individual TFE3 reagents and WT-TFEB rescue, and viability and

rescue expression are rechecked in each background. Persistence under competent combined depletion marks a TFEB/TFE3-independent route; loss under TFE3 depletion indicates family-context dependence. MITF and other routes remain candidates.

3.5 Mechanistic Follow-Up with Defined Decision Rules

AMPK and Akt dependence. WT/3A, both architectures and all three treatments are prespecified for R03; selecting only the largest R02 difference is exploratory. PRKAA1 and PRKAA2 are co-depleted with two independent siRNA pairs in separate experiments, alongside non-targeting, each single-target depletion, no-rescue and mock controls; total siRNA burden is matched, including the concurrent TFEB siRNA. Each catalytic subunit, MK-8722-induced p-ACC, p-S6K, p-AKT(S473), viability, nuclear exposure and rescue abundance are verified at the relevant times. HeLa cells lack LKB1 [8], so AMPK Thr172 phosphorylation relies on other upstream kinases such as CaMKKβ; p-ACC and p-AMPK(T172) confirm MK-8722 engagement, and AMPK conclusions refer to this LKB1-deficient background. If Akt activity differs between the drugs, WT/3A is compared under a pharmacological Akt inhibitor to separate the Akt and AMPK routes at S467. A selective loss of the MK-8722 response with a retained Torin1 response supports stimulus-dependent AMPK involvement, and Torin1 serves as a comparator. If both responses fall, loss of cellular competence and altered baseline expression are evaluated. A near-endogenous, RNAi-resistant AMPK rescue with a matched empty-vector arm gives the target-specific causal test; the required isoform/complex and rescue constructs are sequence-defined at procurement. Without it, the finding is reported as convergent RNAi evidence for involvement. Kinase or phosphatase identity at the C-terminal sites comes from the phosphosite module below.

Phosphosite specificity. In WT tagged and untagged samples, C-terminal phosphorylation is measured together with total TFEB at baseline and after treatment. Phospho-antibodies are tested with phosphatase-treated material, unmodified/phosphorylated peptide controls and site-mutant controls, because adjacent residues can affect epitope recognition. Loss of signal in 3A establishes antibody specificity, and physiological dephosphorylation is shown by a phosphatase-sensitive, treatment-dependent change in WT. If specificity is unresolved, targeted LC–MS/MS with site-localization evidence, peptide standards and total-TFEB normalization is used, and ambiguous adjacent-site assignments are reported as such. Both architectures are enriched through a validated common TFEB epitope outside the altered region, with input/recovery documentation and IgG or bead controls, since GFP-only capture would exclude the untagged arm. Relative phosphopeptide abundance is reported with total-TFEB normalization; absolute site occupancy requires calibration. 3T is analyzed with an assay capable of detecting threonine phosphorylation. This follow-up depends on instrument access and assay development.

Promoter engagement. R05 starts after a qualified RNA-production difference and comparable nuclear exposure are established. TFEB occupancy near validated MCOLN1 and LAMP1 regulatory regions is tested by ChIP–qPCR at matched collection times, with input, matched IgG, TFEB-depleted/no-rescue material, a negative genomic region and a positive locus with demonstrable WT enrichment. A common TFEB antibody outside the changed region is used, with equivalent recovery, chromatin fragmentation and crosslinking conditions validated across architectures. Matched cell equivalents are used and percent input is reported, with a validated spike-in approach if global recovery differs, since percent input does not correct for nuclear TFEB abundance or antibody accessibility. Reduced occupancy supports altered engagement once these checks pass. Comparable occupancy with different nascent RNA motivates cofactor studies. If the checks fail, occupancy is reported as unresolved.

Table 10: Interpretation of possible extension outcomes and the follow-up that resolves each remaining alternative.

Observed pattern, if obtained	Supported interpretation and resolving step
3A defect only in EGFP-tagged TFEB ($A_s \neq 0$)	Fusion-architecture dependence. Linker, RNA, antibody and delivery alternatives are resolved with the R06 linker-only constructs and the transgene RNA and expression checks.
3A defect in both tag states	The effect is independent of the tag; phosphorylation loss and local structural change are the remaining explanations.
3T resembles WT while 3A differs	Local side-chain chemistry contributes; 3T phosphorylation is measured to separate hydroxyl retention from phosphorylation.
Different effects under Torin1 and MK-8722	Stimulus dependence after verified pathway engagement; the AMPK/Akt module identifies the route and the phosphosite module addresses direct modification.
Mature RNA differs but nascent RNA does not	RNA stability/processing contributes; stability studies follow.
Nuclear exposure or abundance also differs	The total effect stands; matched-exposure analyses assess intrinsic transactivation.
No clear interaction with wide intervals	The interval sets the precision achieved; component effects are inspected because equal effects under both drugs cancel in C.

3.6 Linker-Only Constructs for EGFP Attribution (R06)

If R02 shows an architecture effect ($A_s \neq 0$, with an interval excluding zero), two further sequence-defined constructs split it into a linker/stop-position component and an EGFP component (Figure 8, Table 11). TFEB-WT-linker and TFEB-3A-linker encode TFEB residues 1–476 and the GDPPVAT linker followed by a TAA stop. Bp 2080–2799 of the fusion (EGFP including its stop) is replaced by TAA, which gives 5,417 bp and 483 residues. The HindIII and BamHI sites remain, and the HindIII/BamHI digest gives 1,439 + 3,978 bp. Sequence files and the independent audit are in `linker-v4`. The procurement route is whole-plasmid synthesis with whole-plasmid sequence verification, as for the other five constructs, because the Sanger primers S-F5 and S-R2 bind inside EGFP.

Table 11: Linker-only attribution constructs. Coordinates refer to the accepted 6,134-bp WT fusion.

Construct	bp / aa	Intended difference from the matching EGFP fusion
WT, linker-only	5417 / 483	Replace bp 2080–2799 (EGFP including stop) with TAA; GDPPVAT linker, HindIII and BamHI retained.
3A, linker-only	5417 / 483	The same replacement in the 3A fusion, with the three AGC-to-GCC changes.

R06 repeats the R02 primary block with the WT/3A linker-only arms added in the KD background at all three stimuli, which gives 6 added conditions and 51 conditions per day. The linker-only constructs are quantified with the same anti-TFEB reagent, dosed to the same expression window and run under the R02 gates. With L denoting the linker-only architecture, the architecture effect splits as $A_{s,d} = A_{s,d}^{E-L} + A_{s,d}^{L-U}$, where

$$A_{s,d}^{X-Y} = (Y_{3A,X,s,d} - Y_{WT,X,s,d}) - (Y_{3A,Y,s,d} - Y_{WT,Y,s,d}). \tag{5}$$

A^{E-L} is the EGFP component and A^{L-U} the linker/stop-position component. The effect is assigned to EGFP when A^{E-L} differs from zero and A^{L-U} does not, to the linker or stop position in the reverse case,

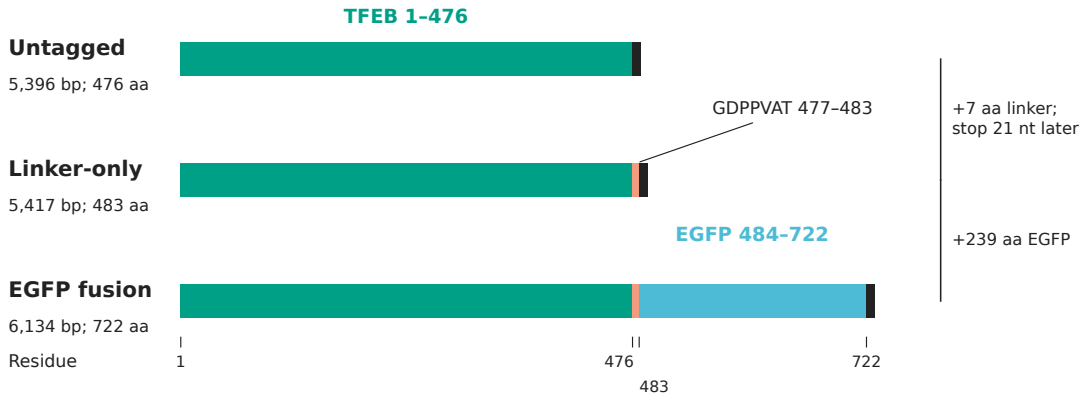


Figure 8: Fusion architectures compared in R06. Untagged TFEB ends at residue 476 with its native stop; the linker-only construct adds the seven-residue GDPPVAT linker (coral) and a TAA stop 21 nt later; the EGFP fusion adds 239 residues of EGFP (cyan). Black blocks mark stop codons, and plasmid lengths and residue counts are read from the sequence files.

and shared between them when both differ. The four contrasts (A^{E-L} and A^{L-U} at each drug) form one Holm family, with sample size set as in R02. Linker-only controls are run only after the R02 architecture effect is established.

3.7 Additional Workload and Feasibility

The extension retains six modules after M01–M12: R01, verify the five constructs and qualify expression/depletion; R02, run the factorial comparison, second-siRNA confirmation and conditional RNA-production/family-context checks; R03, test AMPK and Akt dependence; R04, resolve phosphosite specificity; R05, test promoter occupancy once its entry criteria are met; and R06, attribute an architecture effect to EGFP or the linker with the two linker-only constructs. Weeks 9–12 form a provisional preparation/pilot window. R04–R05 and the conditional R02/R03 additions depend on assay development, resources and preceding results. The eight-week core plan excludes this extension.

At the minimum six confirmatory days and two culture wells per condition, the 18 research conditions alone require 216 wells. Including controls, the primary block requires $45 \times 6 \times 2 = 540$ culture wells, and the second-siRNA WT/3A confirmation requires $39 \times 6 \times 2 = 468$: 1,008 wells *per assay and endpoint* across the two campaigns. R06, when triggered, adds $51 \times 6 \times 2 = 612$ wells per assay and endpoint. These are lower-bound planning counts; they exclude the independent pilot, the six-point destructive imaging series, RNA/protein duplicates on separate plates, additional doses, clone checks, R03–R05 and technical PCR reactions. Scaling to multiwell plates is set by a density/delivery pilot. A physical plate map with all controls is released before execution. If capacity is insufficient, R01/R02 are completed first and the remaining modules are deferred, with all failed and null records retained.

4 Expected Results and Criteria for Success

4.1 Computed Construct and Experimental Predictions

The in-silico pEGFP-N1-TFEB-Bv1 (6,134 bp; Figure 9) encodes a 722-residue TFEB–EGFP fusion: 476 TFEB residues, the seven-residue GDPPVAT junction, and 239 EGFP residues. The native

TFEB stop is absent, and EGFP supplies the final TAA. The calculated unmodified average mass is 80.42 kDa.

The reconstruction and an independent standard-library audit agreed on sequence, reading frame, junctions, terminal stop and fragment geometry. The audit rejected four software challenges: reversed TFEB, a one-base frameshift, restored native TFEB stop, and an internal HindIII site.

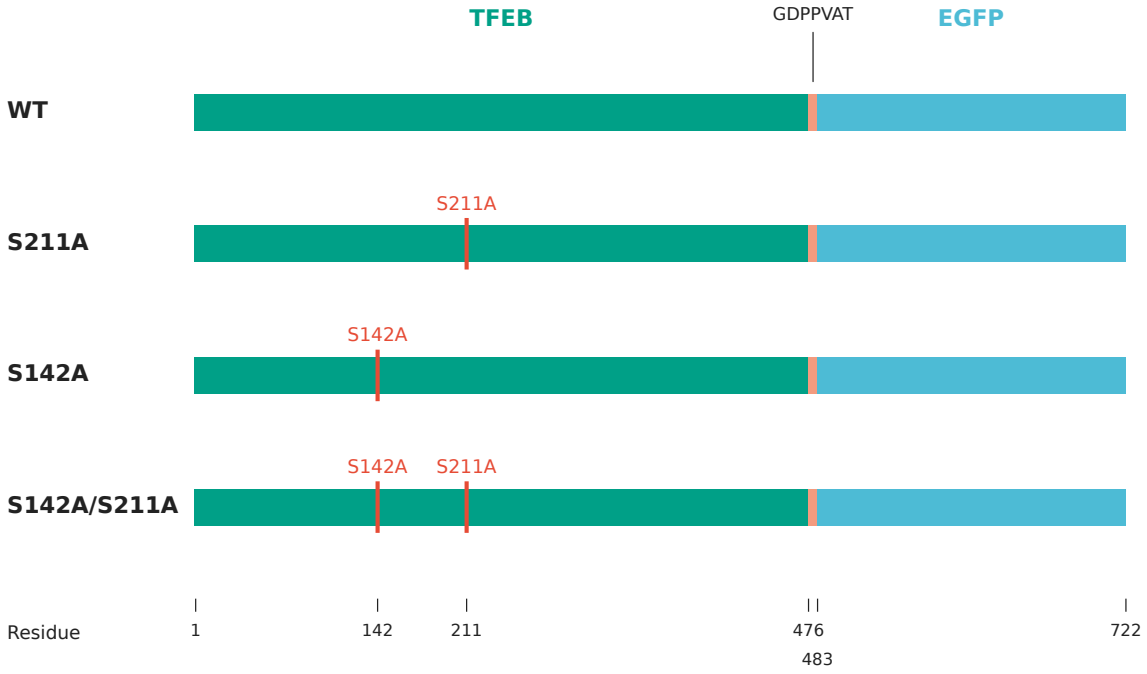


Figure 9: Aligned fusion-protein schematics for WT and three designed mutants. TFEB occupies residues 1–476, the GDPPVAT linker 477–483 and EGFP 484–722; rectangles are proportional to their encoded lengths. Red marks identify S-to-A substitutions. The accepted WT plasmid is pEGFP-N1-TFEB-Bv1 (6,134 bp). The full sequence has HindIII and BamHI recognition starts at 622 and 2061; the first reference-strand bases after cleavage are 623 and 2062. Protein schematics omit regulatory DNA.

Cloning PCR predicts a 1,457 bp product. HindIII/BamHI digestion yields a 1,439 nt insert and 4,695 nt backbone by the cut-to-cut single-strand convention. Their pydna duplex spans are 1,443 and 4,699 bp, including four-base overhangs; gel sizes use the nominal single-strand lengths. Because the empty vector loses a 38 nt MCS segment, the final length is shorter than the intact vector length plus the undigested amplicon.

Table 12 and Figure 11 summarize screening. Flanking PCR distinguishes insert-bearing candidates from empty vector. A negative orientation assay is read together with an amplification-positive control. If all orientation assays are negative in the first round, flanking-positive candidates go to sequencing. Reverse TFEB is a hypothetical challenge; incompatible-end directional ligation does not favor it.

Flanking PCR and HindIII/BamHI digestion do not separate reversed TFEB from the intended construct, and the three-base retained-stop difference falls below gel resolution. Acceptance therefore rests on the expected digest plus actual, unambiguous, concordant bidirectional sequence across 602–2799 (2,198 nt), including both junctions, the complete fusion ORF and terminal stop. Ten primers predict usable coverage in both directions at every target base. Low-quality calls, mixed peaks, gaps and unexpected target changes are resolved under the Sanger acceptance procedure before cell assays. Figure 10 summarizes the planned workflow and controls.

Table 12: Predicted screening products. All sizes are nominal bp; "none" means no specific product is predicted. R and S are hypothetical challenge sequences.

Sample	Flanking PCR (S-F1/S-R1)	Orientation PCR (S-F2/S-R1)	HindIII/BamHI digest
Intended construct (C)	2,605	2,093	4,695 + 1,439
Empty vector (E)	1,204	none	4,695 + 38
Reverse TFEB (R)	2,605	none	4,695 + 1,439
Retained TFEB stop (S)	2,608	2,096	4,695 + 1,442

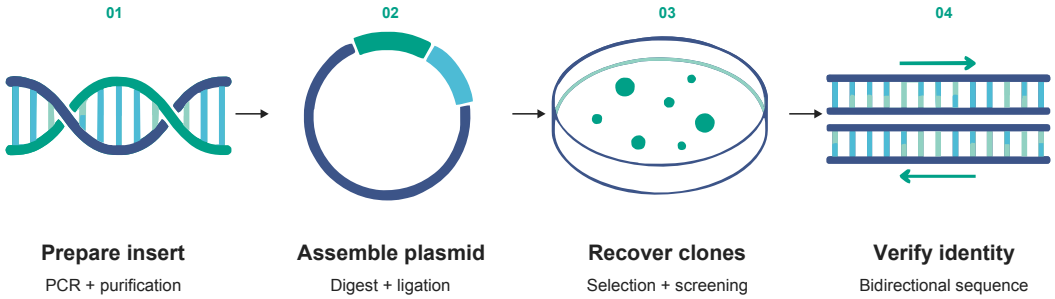


Figure 10: Proposed cloning workflow. Ordered-template identity is checked before PCR; a no-template control, vector-only ligation, no-DNA transformation and a separate competence control accompany the reactions. Backbone and insert are purified after separate digests. Flanking and orientation PCR plus diagnostic digestion screen gross identity. Final acceptance requires unambiguous, concordant bidirectional sequence across the target and both junctions; failed candidates are resequenced or replaced. Measured DNA concentrations set the actual reaction inputs.

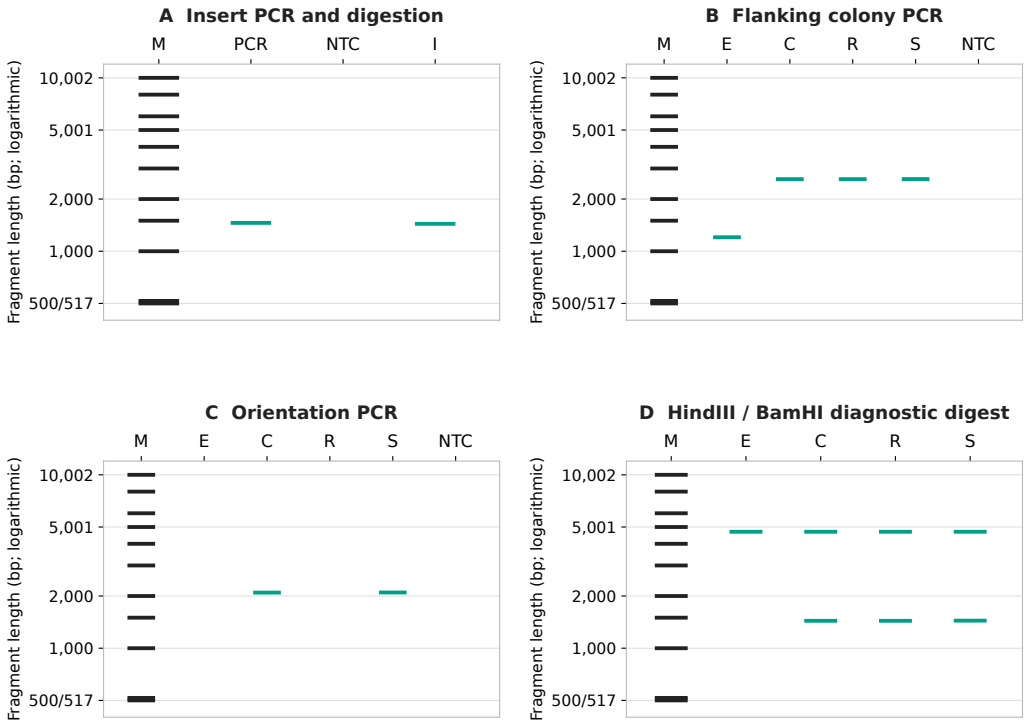


Figure 11: Predicted DNA-band positions, drawn as a schematic of the expected gel. M: NEB N3232 ladder; PCR: tailed insert amplicon; I: digested insert; NTC: no-template control; E: empty vector; C: intended construct; R: hypothetical reverse-TFEB model; S: hypothetical retained-stop model. The 500/517 bp marker pair co-migrates [27]. Panels B and C use the flanking and orientation primer pairs defined in the screening table. The 7, 11 and 38 nt fragments fall below the displayed range. The three-base difference between C and S falls below the resolution of the 1% gel. The empty NTC schematic indicates that no specific product is predicted. Uncut circular DNA mobility and band intensity are not drawn.

4.2 Localization, Mechanism and Function: Decision Matrix

The following are directional expectations and decision rules for each module. The sequence-verified construct is the entry gate for every cell assay.

Table 13: How outcomes would support, limit or challenge the proposed explanation.

Test	Supporting pattern	Alternative result and next check
WT response	Higher N/C after amino-acid withdrawal or Torin1; lower N/C after refeeding than continued starvation	No change: inspect treatment engagement, viability, protein integrity and segmentation before concluding the reporter is unresponsive.
Endogenous agreement	Endogenous IF and clean fractionation show the same directional response as the reporter	Reporter-only response: test abundance dependence and label effects; restrict the finding to the reporter.
S211 contribution	S211A changes baseline distribution and reduces 14-3-3 association relative to matched WT	S211A still responds to Torin1: compatible with multisite regulation. Verify expression and exact sequence if unchanged.
S142 / combined sites	S142A changes recovery relative to WT; double mutant differs from relevant single mutants	No additional double-mutant effect: check dynamic-range ceilings against the leptomycin B arm and starting N/C; redundancy requires a separate test.
mTOR-complex involvement	Independent RPTOR siRNAs reproduce the response with target/pathway engagement; competent RICTOR perturbation differs	Both perturbations act, or RICTOR depletion fails: the attribution is shared between complexes. Resolve cross-talk, toxicity and RNAi specificity.
TFEB-dependent output	Both TFEB siRNAs reduce target-gene responses; WT restores them at controlled expression	Failure of rescue: verify transgene abundance/resistance; residual induction: assess incomplete depletion and TFE3/parallel pathways.
Turnover	Stimulus increases the paired inhibitor-dependent LC3-II increment; loss/rescue alters the TFEB-linked component	High LC3-II without additional inhibitor accumulation points to blocked clearance. Check blockade efficacy, viability and an orthogonal assay.

Localization is concluded when independent-day effects reproduce and agree with the protein and imaging quality controls. A TFEB contribution to target-gene RNA responses is concluded from the loss-and-rescue comparison, and transcription rate from the nascent-RNA assay. Increased operational autophagic turnover is concluded from the paired-inhibitor result. Kinase–substrate chemistry, exclusive TFEB mediation and responses outside the tested HeLa conditions are addressed by the dedicated modules and the second-cell-system confirmation described in the Discussion.

5 Discussion and Scope

The design links construct identity, nutrient-sensitive distribution, selected regulatory sites and downstream function. Its strength comes from comparisons that separate competing explanations:

matched nutrient/solvent controls, endogenous confirmation, expression titration, individual and combined site substitutions, genetic perturbation, and loss-and-rescue. Execution runs from construct acceptance through the cell-assay pilot and localization/recovery confirmation to the mechanism and function modules, and each stage resolves its quality checks before the next begins.

Each control answers a defined question. EGFP-only reports vector and fluorescent-protein effects, and endogenous TFEB measurements test whether the fusion behaves like the native protein. Site substitutions change local chemistry together with phosphorylation potential. RPTOR/RICTOR comparisons attribute the response to an mTOR complex when knockdown is competent, and a separate biochemical experiment is needed to show direct phosphorylation. Recovery of N/C measures net redistribution. TFEB knockdown and rescue establish a TFEB contribution, and the TFE3 controls address the other MiT/TFE-family mediators.

Nuclear accumulation, transcription and autophagic turnover are distinct endpoints. A stress response can move TFEB into nuclei without improving degradation, and drug-induced autophagy also involves TFEB-independent pathways. The functional conclusions therefore rest on their own genetic and inhibitor controls, and when assays disagree the conclusion narrows to the supported level.

The C-terminal extension adds a question beyond reproducing redistribution. Its factorial comparison estimates genotype–fusion-architecture–stimulus effects, and the primary RNA endpoint is the total effect in a calibrated expression system. Removing both linker and EGFP isolates the fusion architecture, and the linker-only constructs of R06 split an architecture effect into its linker and EGFP components. Alanine and threonine substitutions change local chemistry together with phosphorylation, which the 3T arm and the phosphosite assay separate. Nuclear abundance and localization are mediators, and a qualified nascent-RNA assay, TFE3 controls and matched nuclear exposure support an intrinsic-activity interpretation. Phosphosite and ChIP measurements carry their own specificity and recovery requirements. The primary contrast can be zero when both drugs share an effect, so component contrasts are always reported, and a null result with a wide interval reports the precision achieved.

The conclusions apply to TFEB isoform 1, the calibrated expression range, HeLa cells and the stated drug regimens. HeLa cells lack LKB1 [8], so AMPK-related findings refer to that background. A key contrast is repeated in a second human line that expresses LKB1, such as HEK293 or U2OS, with its own expression and pathway qualification before any generalization. The research extension raises the workload well beyond an 18-condition screen, and the calendar yields to assay qualification and the fixed precision-based sample size.

This is an experimental proposal, and no wet-lab clones, images, blots, qPCR measurements or flux data have been acquired. The sequence files define the intended mutant identities. Supplier-specific reagents, siRNA targets, qPCR oligos, antibody performance and final doses/sample size are fixed after the documented pilot validation. These implementation items and the authors' actual contributions are completed before execution or course submission.

Author Contributions

The agreed division of responsibility assigns molecular cloning and construct design (part B) to Siyuan Jiang, and the biological rationale and cell-assay design (part A) to Ziyan Song. With Siyuan Jiang's authorization, AI-assisted integration also expanded part A to include control, mechanism, rescue and functional-assay designs. Each author confirms their actual contributions and reviews the integrated manuscript before submission, and the final statement replaces this planned division. The report

describes a proposed study; no wet-laboratory work was performed.

AI Use Declaration

OpenAI Codex was used for literature checking, sequence-design code, local computational checks, drafting and integration, figure preparation, and LaTeX compilation and layout review. Anthropic Claude (Claude Code, Sonnet 5.5) was used for an independent design review, verification of key citations against PubMed abstracts and full texts, and revision of the text. The circular maps were rendered with SnapGene Viewer from sequence-verified display copies; their English size-unit labels and panel layout were typeset from native SVG exports. The NPG/nrc palette follows the open-source ggsci palette inspired by Nature Reviews Cancer. The cloning and evidence-flow illustrations were generated with the built-in ImageGen tool, corrected after inspection, traced into vector paths with VTracer, and combined with editable labels and arrows. Their visual approach was informed by the Bioicons and D2 GitHub projects, and no third-party icon was copied. Aligned fusion cassettes, the architecture comparison, sampling, the factorial matrix and predicted band positions were drawn by code from declared design choices and accepted sequence/fragment tables. Earlier draft illustrations remain archived. Source images, prompts, calculation inputs and reproducible scripts accompany the design. The authors are responsible for the scientific claims, final parameters and contribution statement.

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